Command-conditioned reduced-order actuator

neural proposal above; explicit hydraulic realization below

desired command

6-step causal history

H_k: 6 × 24 samples · 0.25 s

GRU encoder

2 layers × 128
latent h_k

actuation head

dense output

residual head

dense output

load head

dense output

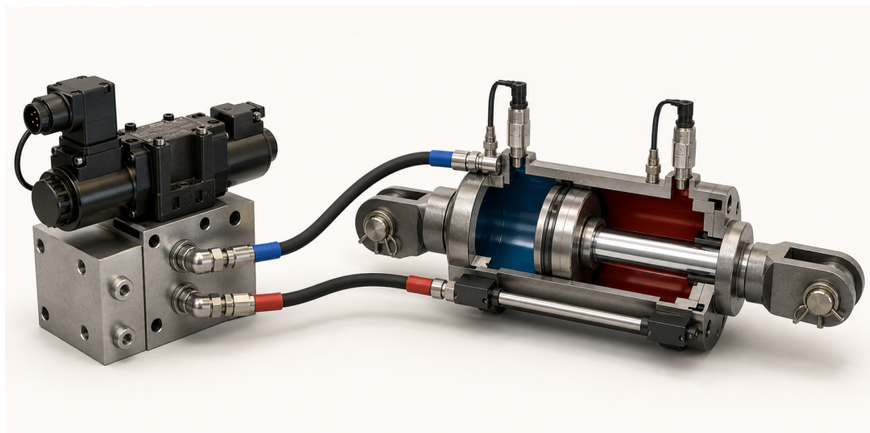
physical proposal

valve · residuals · load

EXPLICIT HYDRAULIC UPDATE

state update

orifice flow → pressure
force → velocity → position



pressure increment $\Delta\hat{p}$

velocity increment $\Delta\hat{x}$

position increment $\Delta\hat{x}$

updated pressure and motion become the next causal history